

Homework 5

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Homework Deliverables

Quick Summary

The goal of this homework is to simulate the static deformation of a clamped rectangular beam under its own weight.

The mesh is made of 20 nodes in 2 rows and 10 columns, with the 4 leftmost nodes fixed. Gravity is the only external force that acts on the beam and viscosity is the only thing dampening the system.

Euler Bernoulli Comparison

The Euler Bernoulli beam theory predicts the tip displacement

$$q = \rho Ag = 0.196 \text{ N/m},$$

where $A = wh$.

The calculated tip displacement is

$$\delta_{EB} = \frac{ql^4}{8YI} = \frac{(0.196)(0.1)^4}{8 \times 10^7 \times (6.667 \times 10^{-12})} = 3.675 \times 10^{-2} \text{ m} = 36.75 \text{ mm},$$

Results

The results from the simulation were a steady state deformation of 29.2 mm from the original position. By comparing the results from the simulation to the Euler Bernoulli model, it gives a normalized difference of 20.55%. This difference which shows a reasonable solution was reached in the simulation.

The figure below shows the change in position over time of the simulated tip displacement. The beam deforms under gravity, and oscillates as it approaches steady state. These oscillations arise from the small viscous damping used in the simulation ($\nu = 0.0001$).

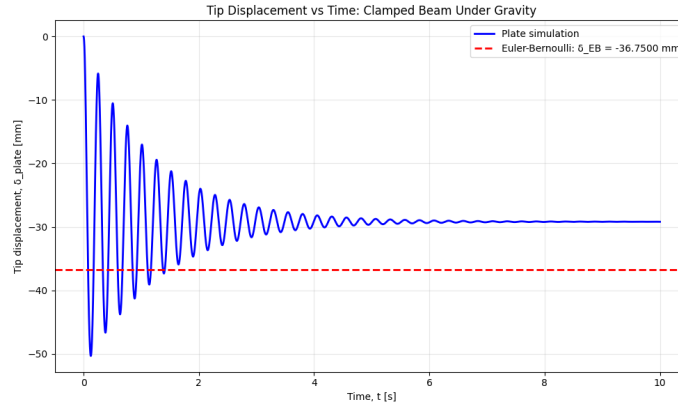


Figure 1: Tip displacement vs. time. Blue: discrete plate simulation. Red dashed line: Euler–Bernoulli prediction ($\delta_{EB} = 36.75$ mm).

Conclusion

The model shows decent agreement with the Euler Bernoulli theory. Some of the ways that the model could be more accurate in the future could be to implement more nodes in the model which would allow for a more accurate solution. Originally the simulation was not producing any oscillations at all but by changing the viscosity from 0.05 to 0.0001, the behavior of the system was shown.

References

- [1] M. K. Jawed, “Lecture 14,” Dept. of Mechanical and Aerospace Engineering, University of California, Los Angeles, 2025.